A mergeable measure of sentiment

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Sentiment scores for text are usually bounded numbers, a class probability or a normalised word count. Such scores do not compose: the score of a chapter cannot be computed from the scores of its sentences, so long texts are cut and averaged, and the answer depends on the cut. Here I describe a measure for which granularity holds. Each word carries positive charge (hedons) and negative charge (dolors) taken from human word ratings, and a span of text is summarised by a signature of nineteen numbers that merges exactly with the signatures of its neighbours. One pass over a book gives the score of every chapter and sentence. Across five corpora with continuous human ratings the calibrated score reaches a mean correlation of 0.51, level with the best lexicon method (0.49) and below fine-tuned transformers (0.59 and 0.68), at 4,500 and 8,700 times their speed on one processor core; on whole film reviews it trails a simpler lexicon method. Two hypotheses from an earlier version were tested against a plan committed before the evaluation data were scored. Feeling words are not front-loaded: they sit late in every kind of text examined, and whether early words weigh more with readers varies by genre. A recency hypothesis that I put in its place passed its three planned tests, but two of them were confounded by the length of the parts compared and reversed once length was held fixed, so I withdraw it. Neutral words dilute a rating, from either side of the feeling. In 60 novels a conventional score moves from 0.11 to 0.76 with the size of the chunks, while merged signatures do not change. Essays that four Claude models wrote to student assignments close on a more positive note than the students' essays and put less of their feeling after intensifiers. A model can be steered to an assigned profile of such measures, and that did not bring its essays closer to the students' than a plain revision did.

Sentiment analysis turns a piece of writing into a number that says how positive or negative it is. The number is computed for product reviews, posts, news, clinical notes and parliamentary speech, often for millions of texts at a time [1–5]. Two families of method are in use. Lexicon methods look each word up in a list of human word ratings and combine the values by rule [6–9]. Classifiers, now mostly fine-tuned transformers, are trained on labelled text and return a class probability [10–12].

Both families return a bounded number for each text: a probability, a mean, or a sum squashed into the interval from −1 to 1. A bounded score is intensive in the sense that temperature is intensive. It does not grow with the amount of text, and the value for a whole cannot be computed from the values of its parts. Three practical problems follow. A transformer reads a few hundred words at a time [13], so a long document is cut into chunks whose scores are averaged, and the average depends on where the cuts fall. A squashed sum saturates: in 60 of the novels studied below, VADER's compound score is beyond ±0.99 for 87% of whole chapters. And an average of averages can rank two bodies of text in the opposite order to their pooled values, the reversal known from Simpson's paradox [14–16]. The third problem is a risk and not an observation: in the data below the two orderings disagree for few pairs, and no comparison reverses.

Here I treat sentiment as a quantity that adds. Every word carries a positive and a negative charge taken from human word ratings. A span of text is summarised by a short vector, its signature, which holds counts and sums of charge weighted by powers of word position. Signatures of adjacent spans merge exactly into the signature of the joined span, so a novel is scored in one pass and every chapter, page and sentence can be read off afterwards without scoring anything again. Scores that people can compare, such as net charge per thousand words or a predicted human rating with an error bar, are read from the signature at whatever scale is wanted.

The parts have precedents. The hedonometer averages human word ratings [2,6]. Word-count tools and studies of mood keep positive and negative words apart [3,9,17], in line with evidence that the two are separable in evaluation [18]. Political scientists score texts from counts of opposed items on a log-ratio scale with a stated uncertainty [4,19]. Summaries that merge exactly across partitions, including sums weighted by polynomial or exponential functions of time, are standard in stream processing [20–23]. Emotional arcs of stories have been drawn with moving windows, transforms and fitted polynomials [24–26]. This paper joins these into one object for sentiment, states which position weights such an object can carry, and tests it against human ratings.

The aim is a mathematical measure of sentiment for which granularity holds: the score of a whole is the merge of the scores of its parts, at every level from the phrase to the book. The work began with a manuscript I wrote in 2024, which proposed that feeling in text could be scored as a number from word values, a decay over position and a penalty for words that carry no feeling. It made two claims about how people write and read. Front-loading: words that carry feeling come early in a text and count for more there. Dilution: words that carry none weaken the feeling of the whole. I wrote both down as hypotheses with decision rules, and committed them to the project repository, before scoring any evaluation data (Methods).

Front-loading fails as a claim about placement: charged words sit late, not early, in every kind of text examined. As a claim about weight it depends on the genre. Readers of film-review sentences weigh the early part more, and readers of tweets and of Yelp reviews the late part. Midway through the work I replaced front-loading with a hypothesis of recency, that what comes last counts most, and committed three tests of it. All three passed by their stated rules. Two of them compared parts of unequal length, and when a referee agent pointed this out and I held length fixed, both reversed. I report the sequence in full and withdraw the hypothesis. Dilution holds, by the planned test in 4 of six corpora and in human phrase ratings on either side of the feeling. A last experiment applies the measure to a current question and finds that essays written by language models differ from student essays on the same assignments in where and how they carry feeling.

Results

A signature that merges

Take a text as a sequence of words. Word i has a positive charge h_i and a negative charge d_i, each between 0 and 1, at most one of them

nonzero. The values come from two open sets of human word ratings, the labMT list[2] (10,222 words, 50 raters each) and the VADER list[7] (7,494 entries, 10 raters each). On the 1,366 words they share, the two sets correlate at $r = 0.95$. I put both on one scale, on which 1 is the positive end of the rating scale and 0 its neutral point, and average them (Methods). Positive charge is measured in hedons (hd) and negative charge in dolors (dl), the names that moral philosophy uses for units of pleasure and pain[27,28]. One hedon is the charge of a word that every rater places at the top of the scale; *love* carries 0.85 hd and *terrible* 0.55 dl. A word rated within a quarter of the scale from neutral gets no charge and is called a buffer word, following the neutral band of the hedonometer[2]. In the six rated corpora used here, 23% of running words are charged.

The signature of a span holds the number of words L, the number of charged words E, and the sums

$$S_k^+ = \sum_i h_i \, i^k, \qquad S_k^- = \sum_i d_i \, i^k, \qquad k = 0, \ldots, 3, \tag{1}$$

where i is the position of the word in the span, counted from zero. S_0^+ and S_0^- are the span's total hedons and dolors. The signature also holds the hedons and dolors that fall under a negation, after an intensifier, after a diminisher, and after a contrast word such as *but*, together with the sum of squared charges (Methods). That is 19 numbers, whatever the length of the span (Fig. 1a,b).

Signatures merge. If span A is followed by span B, the counts and totals add, and the position sums of B are shifted by the length of A:

$$S_k(AB) = S_k(A) + \sum_{j=0}^{k} \binom{k}{j} L_A^{\,k-j} \, S_j(B). \tag{2}$$

The operation is associative and can be undone, so the signature of B follows from those of AB and A (Proposition 1, Methods). A text of any length is therefore scored in one pass with constant memory. Parts can be scored separately and merged in reading order to the same result as a single pass. A table of prefix signatures, one row per sentence, gives the signature of any contiguous run of sentences from two of its rows. In integer arithmetic these statements hold to the last digit, and the reference implementation, which counts in thousandths of a hedon, is tested for exact equality on random texts. The fast implementation uses double-precision floats; over 356 novels its merged and single-pass signatures agree to a relative error below 2×10^{-13}.

Which position weights allow a merge? Suppose a summary keeps finitely many sums $\sum_i c_i \, \varphi(i)$ of charge against weight functions φ. The summary of AB can be computed from the summaries of A and B and their lengths exactly when the weight functions span a space that is closed under shifting the position, and such weights are combinations of $i^k \rho^i$: polynomials, exponentials and their products (Proposition 2, Methods). The statement is an instance of a classical fact about translation-invariant function spaces[29,30] and about functions on strings with finite-dimensional representations[31,32], and stream-processing systems rely on it[20]. For sentiment it says two things. The decay factor of the 2024 manuscript, a weight that falls geometrically with position, is one of the few position weights that survive a merge, at the cost of one number per decay rate. And among mergeable weights, polynomials alone can be re-expressed in relative position, that is, position divided by a length that is unknown until the text ends. The signature keeps polynomial sums for that reason: the sum of charge under any cubic function of relative position can be read from it, for a post and for a novel alike.

Reading a score from a signature

A signature is extensive. Scores that can be compared between texts are read from it (Fig. 1c).

Totals are the hedons $H = S_0^+$ and dolors $D = S_0^-$ over L words. *Density* is the net charge per thousand words, $1000\,(H - D)/L$; the hedonometer's average happiness is a density taken over charged words only.

Level follows a model from the psychology of judgement. In Anderson's averaging model a judgement is a weighted mean of the pieces of information received and an initial neutral impression[33], and information that says nothing pulls a judgement toward neutral, the dilution effect[34]. For words this gives

$$\text{level} = 1000 \, \frac{H - D}{a + E + \kappa B}, \tag{3}$$

where $B = L - E$ counts buffer words, a is the weight of the neutral first impression in units of charged words, and κ is the weight of a buffer word relative to a charged one. With $\kappa = 0$ buffer words do nothing; with $a = 0$ and $\kappa = 1$ the level is the density. The denominator is the exposure, the amount of evidence behind the score, and it gives the level a standard error that shrinks as a text grows (Methods).

Calibrated valence is a predicted human rating. It is a linear combination of ten signature entries, each divided by the exposure: hedons, dolors, three position moments of net charge, and the charge under each operator. Given a and κ, the coefficients solve a ridge regression against human ratings, a convex problem with a closed-form solution; a and κ are chosen on a grid (Methods). The fitted coefficients say how much a negation flips a word and how position is weighed; they are estimates with intervals, in the tradition of rule constants measured from ratings[7,35].

The *arc* is the least-squares cubic fit of net charge against relative position. It has a closed form in the position sums (Methods), so it costs nothing beyond the signature.

Agreement with human ratings

I compared scores with human ratings on six corpora: sentences from six genres (EmoBank[36]), tweets (SemEval-2018[37]), Facebook posts[38], film-review sentences (the Stanford Sentiment Treebank, SST[39]), whole film reviews with the critic's rating[40], and Yelp reviews with star ratings[41]. The first five carry continuous ratings and form the primary set. Every constant was fitted leave-one-corpus-out: for each target corpus, a, κ and the ten coefficients came from the other five, and scores were then computed on the target's held-out test portion (Methods). The six fits chose a of 2 or 4 and κ of 0.2 or 0.5.

Averaged over the five primary corpora, the calibrated score correlates with human ratings at $r = 0.51$, against 0.49 for VADER[7], 0.44 for TextBlob and 0.42 for the hedonometer[2] (Fig. 2a; all values with intervals in Extended Data Table 1). The plan set one criterion, which an earlier version of the method had failed: a mean r at least that of VADER (Methods). It is met, 0.506 against 0.489, by a margin that is within noise. The mean difference is +0.016 with a t interval over the five corpora of [−0.017, +0.049], and on the plan's secondary outcome, Spearman's ρ, the two are level (0.506 and 0.508; Extended Data Table 2). Corpus by corpus, the paired difference from VADER excludes zero in favour of the signature in 3 of six corpora and in favour of VADER in 0. On whole film reviews, the longest rated texts, TextBlob does better than both ($r = 0.46$ against 0.29 for the signature; difference −0.168 [−0.200, −0.136]). The level of equation (3), whose only fitted quantities are its own a and κ (1 or 2 and 0.2), reaches 0.48, below VADER.

Methods given more labelled text do better. A bag-of-words ridge regression trained on the same five-corpus pool as the signature's

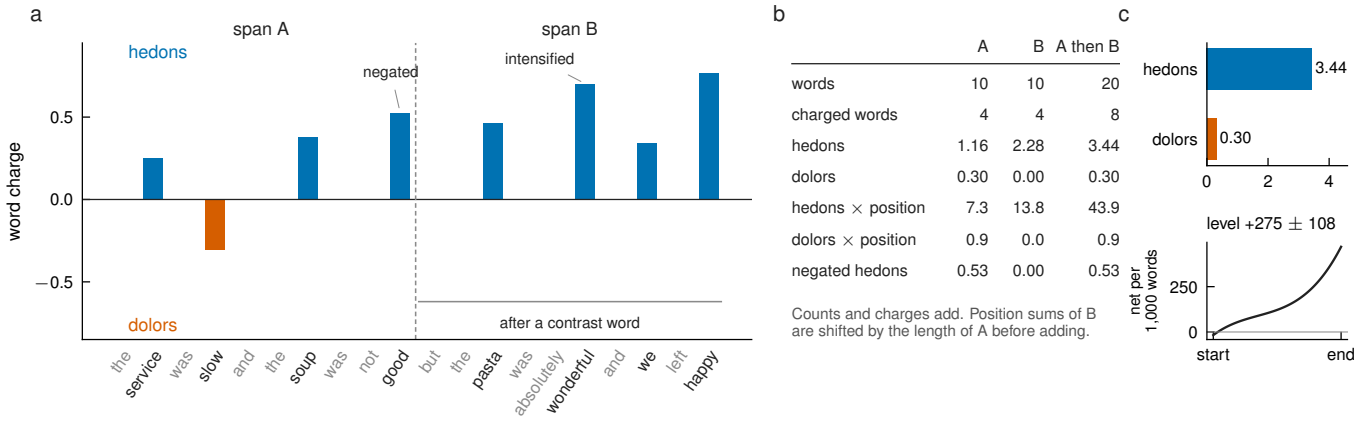


Figure 1 Charge, signature, merge, readouts. a, Two sentences scored word by word. Bars above the line are hedons, bars below are dolors; grey words are buffer words and carry no charge. *Good* sits inside the reach of *not*, *wonderful* follows an intensifier, and the second sentence follows a contrast word. **b**, Entries of the signature of each sentence and of the two read as one text, obtained by merging and equal to a single pass over both. **c**, Readouts of the merged signature: totals, the level with its standard error (the level's own constants, $a = 1$ and $\kappa = 0.2$), and the cubic arc of net charge over relative position.

constants averages 0.55, and trained on each corpus's own training portion 0.61. DistilBERT fine-tuned on SST-2 [12] averages 0.59 and a RoBERTa model fine-tuned on tweets [42] 0.68. The lead of the transformers is uneven. The signature is ahead of DistilBERT on tweets and on Facebook posts (differences +0.103 [+0.062, +0.145] and +0.077 [+0.036, +0.115]), level with it on EmoBank, and far behind on SST, the domain DistilBERT was tuned on, and on reviews. The signature is not a replacement for these models where accuracy on one text matters most.

Removing parts of the signature shows what earns its place (Fig. 2c; means over all six corpora). Without the operator channels the mean correlation falls by 0.034, and without dilution ($\kappa = 0$) by 0.015. Without the neutral prior ($a = 0$) it falls by 0.010, a mean over a gain from the prior in 2 corpora, a small loss in film reviews and no clear change in the other three. Merging hedons and dolors into one channel costs 0.002 and dropping the position moments 0.000: neither matters for agreement with raters, though both are needed for the readouts below.

The fitted coefficients can be read as rules, with a caution: the readings belong to one pooled fit, the shipped model, and the bootstrap intervals describe sampling error within that fit and say nothing about other corpora. In it a dolor moves the predicted rating 1.63 [1.54, 1.73] times as far as a hedon. A negated positive word counts as −1.60 [−1.85, −1.37] times its charge and a negated negative word as −0.01 [−0.23, +0.20]: *not good* is bad, and *not bad* is neutral. An intensifier adds +1.02 [+0.85, +1.19] times the charge of the word it modifies. Across the six leave-one-corpus-out fits the first three readings run from 1.17 to 1.81, from −1.95 to −1.10 and from −0.14 to +0.10, and fits to single corpora differ far more (Extended Data Table 5): the ratio of dolor to hedon is below 1 in SST and above 4 in Yelp. One reading has the same sign in every fit, pooled or single: a negated positive word turns negative. The sizes are properties of these corpora. The data do not settle the effect of diminishers (−0.33 [−0.91, +0.09]) or of contrast words (−0.06 [−0.14, +0.02]); in particular they give no support to the rule that the clause after *but* counts for more [7].

Word charges can also be fitted. With the exposure and the operator constants held fixed, the predicted rating is linear in the vector of charges, so the charges that best predict ratings, pulled toward the human word norms by a ridge penalty, solve one convex problem (Methods). Fitted leave-one-corpus-out, these refined charges raise the mean correlation to 0.55, the same as the bag-of-words ridge trained on the same five corpora (0.55). The gain is uneven. It is large on review text (SST 0.44 to 0.55, film reviews 0.29 to 0.49) and negative on the mixed genres of EmoBank (0.48 to 0.41) and on Facebook posts (0.62 to 0.58). Charges learned from ratings carry the genre of the texts they were learned from, and their attitudes. Of 20 words for groups of people and two public figures, checked after a referee agent raised the question, 14 moved down by more than 0.05 on the charge scale (*women* from +0.53 to −0.08, *muslim* from −0.09 to −0.37; Extended Data Table 7). Human word norms travel better, and the refined charges are not part of the released package.

One hoped-for property did not hold. The standard error of a level shrinks with exposure, and the error of the calibrated score against human raters does not follow it. From the lowest to the highest quartile of exposure the error falls in 2 corpora (EmoBank and Yelp) and rises in the other 4 (Extended Data Fig. 1a). The standard error says how well a text's own words pin down its level. It is not a prediction interval for a human rating.

Cost

Scoring is one table lookup per word. On one core of a rented six-core desktop processor the implementation, written in Python with NumPy, turns 167,000 sentences per second, or 2.6 million words, into signatures and levels (Fig. 2b). On the same core VADER scores 27,900 sentences per second, TextBlob 11,100, DistilBERT 37 and the RoBERTa model 19. The signature is therefore about 4,500 times faster than DistilBERT and 8,700 times faster than the RoBERTa model, well past the factor of 100 that the plan set as its criterion. On a graphics card (RTX 3090, half precision) DistilBERT reaches 12,200 sentences per second, and one processor core running the signature is still 14 times faster. Time grows in proportion to the amount of text (log-log slope 1.02 from 10^4 to 10^6 sentences); ten million sentences took 59 seconds in fixed-size batches. Because signatures merge, the work divides among processes without changing any result, and twelve processes on the six cores scored 0.8 million sentences per second, 4.8 times the rate of one. At these rates a billion sentences take 1.7 hours on one core or 21 minutes on the six, against 23 hours for DistilBERT on the graphics card and 310 days on one core.

Where the feeling sits

Front-loading makes three predictions, and I tested each (Fig. 3).

The first is that charged words come early. They do not. The charge-weighted mean position, on a scale from −1 at the first word to +1

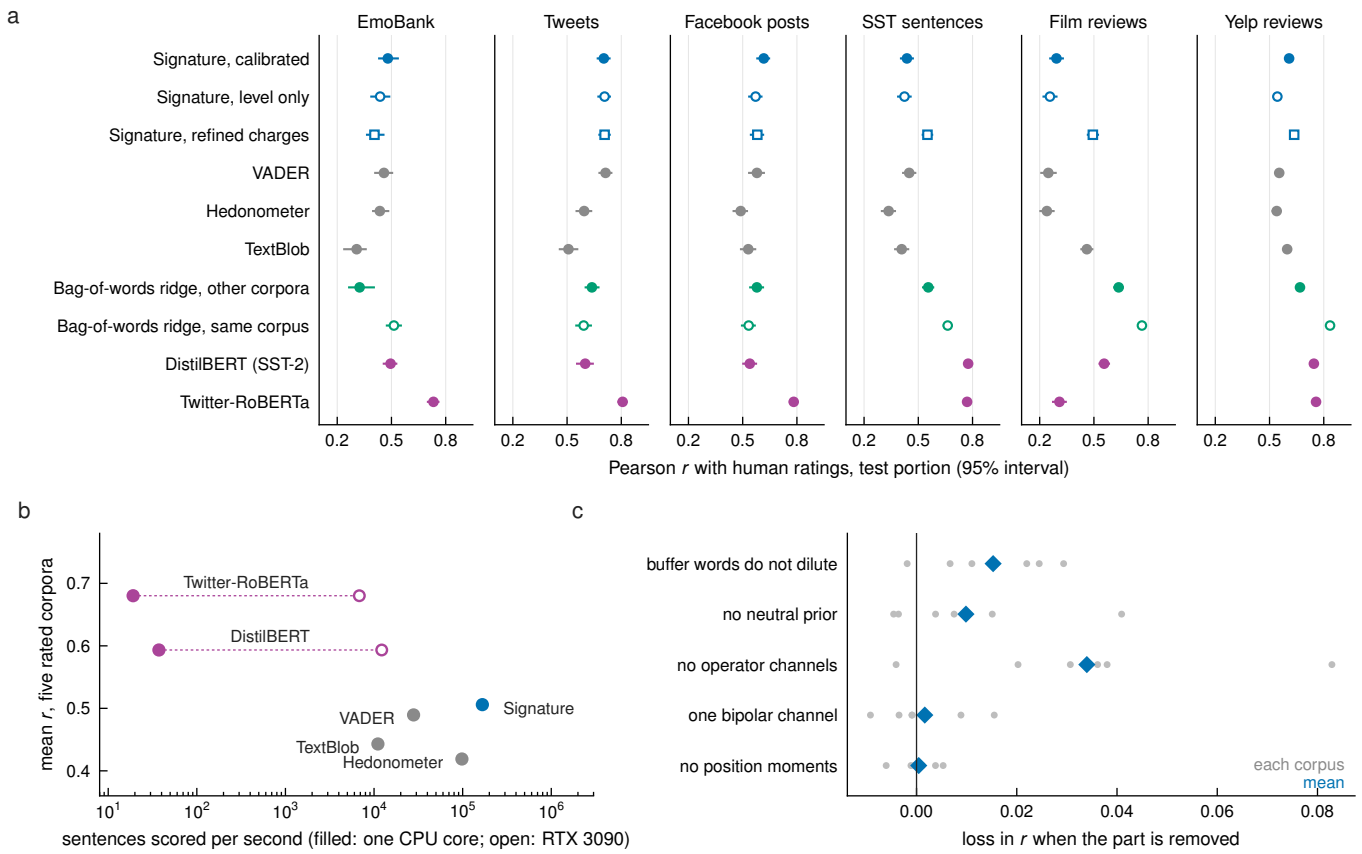


Figure 2 Agreement with human ratings, and cost. a, Pearson r between score and human rating on the held-out test portion of each corpus, with 95% bootstrap intervals. Blue: readouts of the signature, with constants fitted on the other five corpora (filled circle, calibrated valence; open circle, level; square, calibrated valence with refined word charges). Grey: lexicon methods. Green: bag-of-words ridge regression trained on the other five corpora (filled) or on the same corpus (open). Purple: transformer classifiers. **b**, Mean r over the five corpora with continuous human ratings against sentences scored per second on one CPU core (filled) and, for the transformers, on a graphics card (open). **c**, Loss in r when one part of the signature is removed and the constants are refitted.

at the last, is above zero in all 8 kinds of unit examined (the six rated corpora, paragraphs of the film reviews and chapters of 150 of the novels), from +0.002 in Yelp reviews to +0.105 in tweets, with every interval excluding zero (Extended Data Table 3). In EmoBank sentences, SST sentences and tweets the first tenth of the words carries 0.65 to 0.75 of the text's mean charge per word, and in those three and in Facebook posts the last tenth carries 1.34 to 1.55 (Fig. 3a). English sentences tend to open with determiners and pronouns and close on content. Facebook posts open at the mean (1.02). Yelp reviews are the one place with a charged opening, at 1.09 times the mean, matched by the close at 1.11. Negative charge sits later than positive charge in 7 of the 8.

The second prediction is that readers weigh early words more. Here the answer depends on the kind of text. The position kernel fitted in each corpus puts more weight on the first third of a text than on the last third in SST sentences (difference +0.26 [+0.14, +0.39]), as front-loading predicts. It puts less in tweets (−0.39 [−0.54, −0.24]) and in Yelp reviews (−0.12 [−0.15, −0.09]), and is undecided in the other three (Fig. 3c). In the shipped model, fitted on all six corpora, the last third outweighs the first (−0.13 [−0.20, −0.07]).

A third analysis, added after the first results, uses human ratings alone. The SST gives a human rating to every phrase of its parse trees[39], so the rating of a phrase can be regressed on the ratings of its two parts. In the training trees the later part gets the larger weight, 0.68 against 0.61 where both parts have at least two words (29,636 phrase pairs). The comparison is confounded by length. The later part is the longer one in 62% of pairs (7.8 words on average against 4.7), and the weight of a part rises with its share of the phrase's words, from 0.29 for a part with under a tenth of them to 0.86 for a part with over nine tenths (Fig. 3b; all trees, 41,110 pairs). Where the two parts are equally long, the earlier one has the larger weight: 0.71 against 0.57 (later minus earlier, −0.138 [−0.182, −0.090]; 5,366 pairs). With one curve of weight against word share for both parts, the extra weight of the later part is −0.019 [−0.035, −0.002]. So in film-review phrases the longer part counts for more, and length for length the earlier part counts for slightly more. English syntax may be the cause as much as order, since the two parts of a phrase differ in grammatical role. The dependence on length is modest: a plain mean of the two part ratings accounts for the phrase rating as well as a mean weighted by length (R^2 = 0.59 and 0.58; 0.59 with free weights).

The plan's third test, within EmoBank documents, is undecided: rated intensity neither falls nor rises with sentence position (slope +0.012 [−0.012, +0.032] per document length; 115 documents).

Front-loading is therefore refuted as a claim about where writers put feeling words. As a claim about what readers weigh it holds in film-review sentences, by the kernel and by the phrase ratings at equal length, fails in tweets and in Yelp reviews, and is undecided elsewhere. For whole reviews, earlier work found that the closing sentences carry the verdict[43–46].

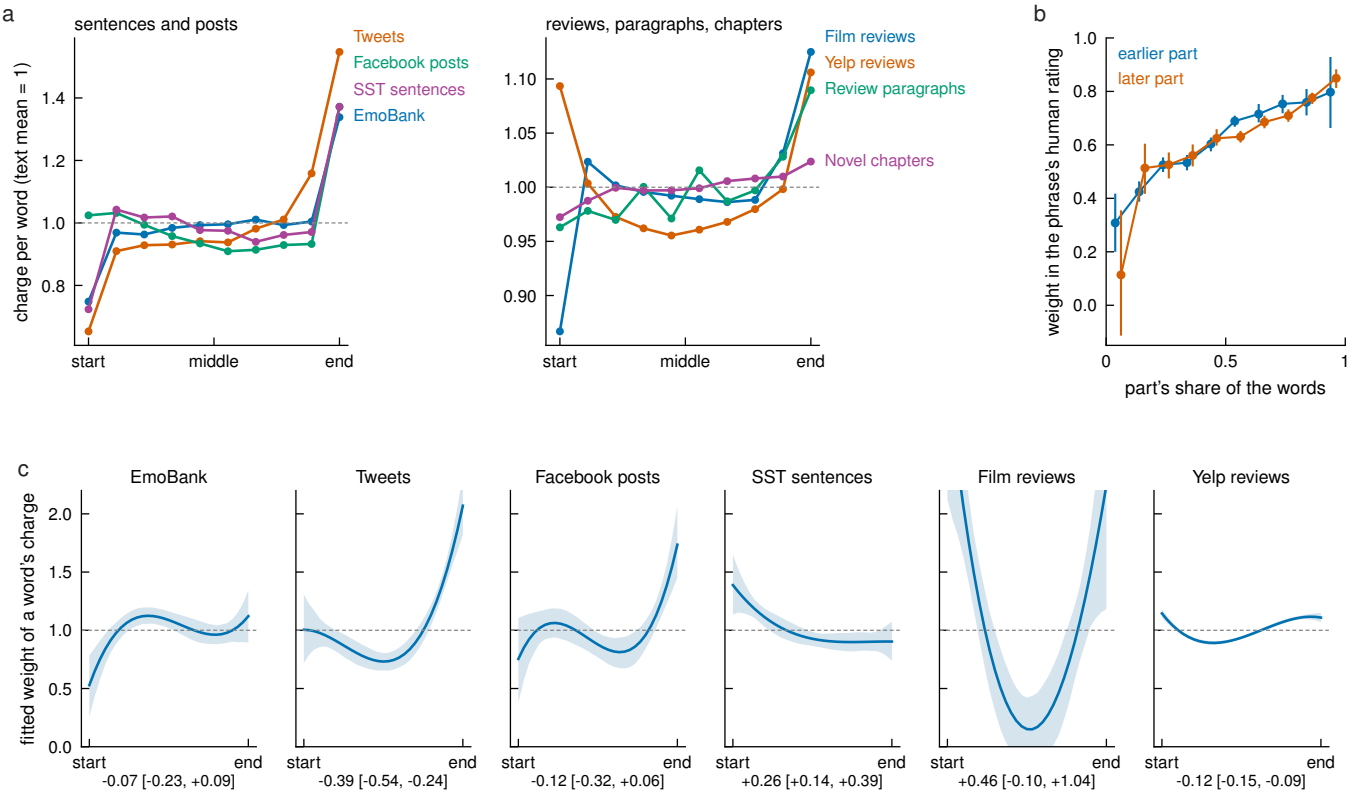


Figure 3 Where feeling sits, and what readers weigh. a, Charge per word in each tenth of a text, relative to the text's mean, for texts of ten or more words. **b**, Weight of a part in a regression of a phrase's human rating on the human ratings of its two parts, by the part's share of the phrase's words, for earlier and later parts of 41,110 binary nodes of the Stanford Sentiment Treebank (95% intervals, sentences resampled). Weight rises with share for both, and at equal share neither part leads consistently. **c**, Position kernel fitted against human ratings in each corpus: the weight given to a word's charge as a function of its relative position, with 95% bands. Below each panel, the mean weight over the first third minus the mean over the last third. The kernel for whole film reviews is poorly determined.

Neutral words dilute, from either side

Dilution fares better (Fig. 4). The planned test asks whether a readout with a buffer weight $\kappa > 0$ agrees with held-out ratings better than the same readout with $\kappa = 0$. It does in 4 of six corpora (tweets, Facebook posts, SST and Yelp) and is undecided in EmoBank (+0.007 [−0.003, +0.020]) and in film reviews (−0.002 [−0.009, +0.004]). Two statistics that the plan did not list agree in direction. Holding a text's total charge fixed, rated intensity falls as the number of buffer words rises in 5 of six corpora, EmoBank among them, with partial rank correlations from −0.27 in EmoBank to −0.08 in SST sentences (Fig. 4b); whole film reviews show no effect (+0.01 [−0.02, +0.04]). In fits to single corpora, agreement is highest at a buffer weight between 0.2 and 1 in the same five and at zero for film reviews, with gains in r of 0.009 to 0.020 that carry no intervals (Fig. 4c).

Human phrase ratings show the effect without any scorer. Take one rated word of the treebank joined to a neutral phrase of m words. With the neutral words after it, the joined phrase keeps 0.61 [0.60, 0.63] of the word's rating for $m = 1$ and 0.39 [0.27, 0.49] for m of nine or more. With the neutral words before it, the share is 0.54 [0.52, 0.55] for $m = 1$ and 0.16 [0.00, 0.30] for m of 5 to 8 (Fig. 4a). The share falls with m on both sides. Over rated parts of every length, matched on the lengths of both parts, the share kept is −0.050 [−0.058, −0.042] lower when the neutral words come first than when they follow. The level of these shares is pulled down by regression to the mean, because the rated part was selected for a strong rating; the trend with m and the comparison between sides are not.

The 2024 manuscript had dilution right. Words that carry no feeling weaken the judged feeling of the whole, more of them weaken it more, and it does not take a particular order, as the averaging account of impression formation predicts [33,34].

Recency, proposed and withdrawn

Two of the first results, charged words sitting late and the later part of a treebank phrase carrying the larger weight, suggested a hypothesis opposite to front-loading: recency, that what comes last in a text counts most toward a reader's judgement of it. I wrote it down with three predictions and committed them before examining the data they concern, the development and test trees of the treebank and 22,001 IMDB film reviews (Methods). By their stated rules all three held. In 11,474 held-out phrase pairs the later part has weight 0.69 and the earlier part 0.63 (R1; difference +0.059 [+0.032, +0.085]). A rated phrase keeps 0.68 of its rating when two or more neutral words follow it and 0.77 when they come before it (R2; difference +0.091 [+0.055, +0.127]). And the last fifth of an IMDB review separates favourable from unfavourable reviews better than the first fifth does (R3; area under the curve 0.696 against 0.657; difference +0.039 [+0.031, +0.048]).

A referee agent then pointed out that R1 and R2 do not hold length fixed. With length fixed, both fail. In the held-out trees the later part is the longer one in 63% of pairs. At equal length the difference in R1 reverses (−0.164 [−0.257, −0.081]), and with a common curve of weight against word share the later part's extra weight is −0.030 [−0.059, −0.002]. In R2 the rated parts that were followed by neutral words averaged 3.8 words and those preceded by them 9.1, and a short rated part keeps less of its rating whichever side the neutral words are on. Matched on both lengths the difference reverses (before minus

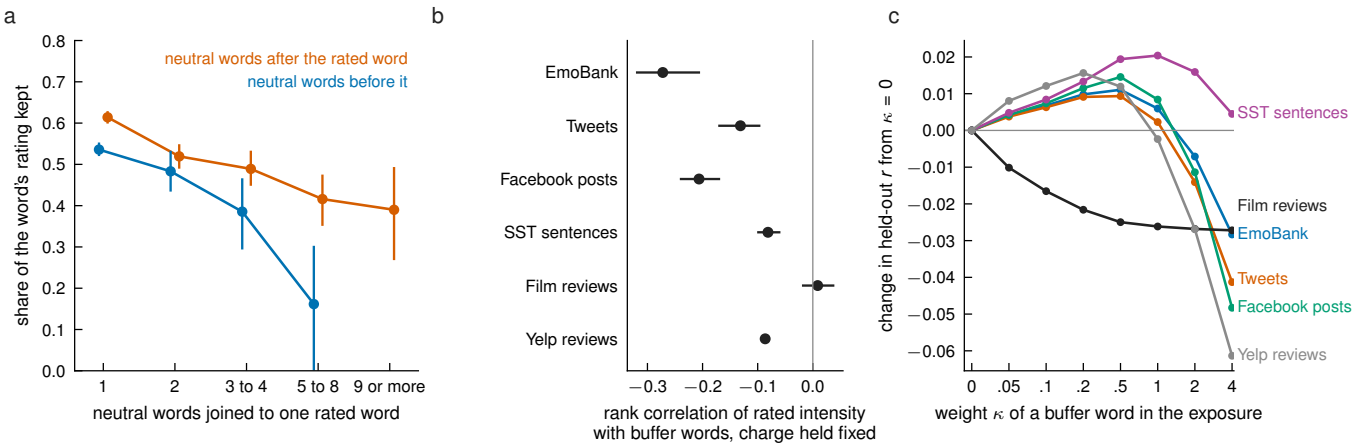


Figure 4 Neutral words dilute a rating from either side. a, Human phrase ratings from the Stanford Sentiment Treebank. One word with a clear rating is joined to a neutral phrase; the vertical axis is the share of the word's rating that the joined phrase keeps, by the number of neutral words and their side (95% intervals; cells of 30 or more phrases). **b**, Partial rank correlation between rated intensity and the number of buffer words in a text, with total charge held fixed. **c**, Change in held-out r as the weight κ of a buffer word in the exposure moves away from zero, in fits to each corpus alone with a fixed at the shipped value and coefficients refitted at each κ; the grid is extended to $\kappa = 4$.

after, −0.050 [−0.064, −0.035]).

R3 compares fifths of equal length and stands. It measures where reviewers place their verdict, which is a fact about writers. The scorer's density in the first, the three middle and the last fifth separates the two classes with areas of 0.657, 0.648 to 0.651 and 0.696: the verdict shows at both ends of a review and most at the end, as the earlier work on reviews found[43,46].

Recency as a claim about readers is withdrawn. Its tests were written down in advance and passed, and they were the wrong tests, because the plan left out the control they needed. Both steps are on record in the analysis plan.

A book at every scale

I scored 356 novels from Project Gutenberg (28.1 million words, median 25 chapters) sentence by sentence and merged the signatures up to chapters and books. Fig. 5a,b shows *Moby-Dick*: 213,523 words carrying 11,790 hd and 3,539 dl, a density of +39 per thousand words, with each of the 129 sections that the chapter rule finds in it (the book has 135 chapters and an epilogue; Methods) and twelve spans of the last one read from the same pass.

Chapters are not random samples of a book's sentences. The variance of chapter densities is 5.9 times (median; interquartile range 4.3 to 7.8) what results when a novel's sentences are shuffled across its chapters, and exceeds the shuffled value at $p \leq 0.01$ in 98% of novels (Fig. 5c). The shuffle destroys all continuity between neighbouring sentences, so the ratio shows that tone is locally coherent in novels. It does not show that chapter boundaries are where tone changes, which would need a null that moves the boundaries and keeps the order.

The cubic arc read from a book's signature captures 41% (median) of the variance of an arc drawn with a moving window one tenth of the book wide; keeping moments up to degree nine would capture 63% (Fig. 5d). The two leading modes of variation among the window arcs of the 356 novels carry 22% and 19% of the variance and are close to the first two Legendre polynomials (cosine similarity 0.94 and 0.89), the rise or fall and the single turn that earlier studies of story shapes found[24]; the third mode is not close to the cubic (0.10). A wide moving window yields smooth leading modes from any series, so the resemblance is weak evidence about stories.

Human labels exist for one body of narrative. Alm and colleagues labelled every sentence of 176 fairy tales with an emotion[47]. The cubic arc of each tale's merged signature agrees in sign with the cubic arc of its human labels in 70% of tales (median correlation 0.46; sign test $p = 6 \times 10^{-8}$; Extended Data Fig. 1b). For a tale's overall valence the picture is less favourable. The calibrated score of the merged signature correlates with the mean human label at 0.50 [0.34, 0.63], and VADER averaged over sentences does better (0.63 [0.51, 0.73]). The human target here is itself a mean of sentence labels, which an average of sentence scores mirrors. VADER applied to the whole tale saturates beyond ±0.99 for 73% of tales and falls to 0.39.

A caution applies to every number in this section. Word norms rate words, not events. The final chapter of *Moby-Dick*, in which the ship is destroyed, scores near the book's own level, because *whale*, *sea* and *boat* are pleasant words to raters. Densities of narrative text are best read against the text's own baseline.

Pooling, averaging and chunking

Bounded scores depend on how a long text is cut (Fig. 6a to c). For the first 60 novels in file order I computed a book score as the mean VADER compound score over chunks. The mean across books rises from 0.11 with one-sentence chunks to 0.76 with whole chapters, and 19% of pairs of books swap order between the two. The same happens on 300 whole film reviews. The score of a review and the mean score of its sentences differ in sign for 28% of reviews with DistilBERT, 51% with the RoBERTa model and 5% with VADER, whose whole-review score is beyond ±0.99 for 48% of them. For the transformers the comparison is confounded by truncation: 97% of the reviews exceed the models' 512-token window, so the whole-review score reads the opening of a review and the sentence mean reads all of it. In all three cases the mean of sentence scores predicts the critic's rating better than the whole-review score does ($r = 0.61$ against 0.53, 0.52 against 0.23, and 0.23 against 0.17).

What depends on the cut is the average. If the densities of chunks are averaged, 8% of pairs of books swap order between one-sentence chunks and chunks of 125 sentences, against 11% for VADER (10% and 19% with whole chapters). If the chunks' signatures are merged, the book's density is the same for every chunking, to a relative difference below 2×10^{-14}.

Invariance is not validity. On the same 300 reviews the merged density correlates with the critic's rating at 0.23, far below the mean of DistilBERT's sentence scores (0.61), and among the fairy tales an average of VADER's sentence scores tracked overall valence better than the merged signature did. Merging removes the dependence on

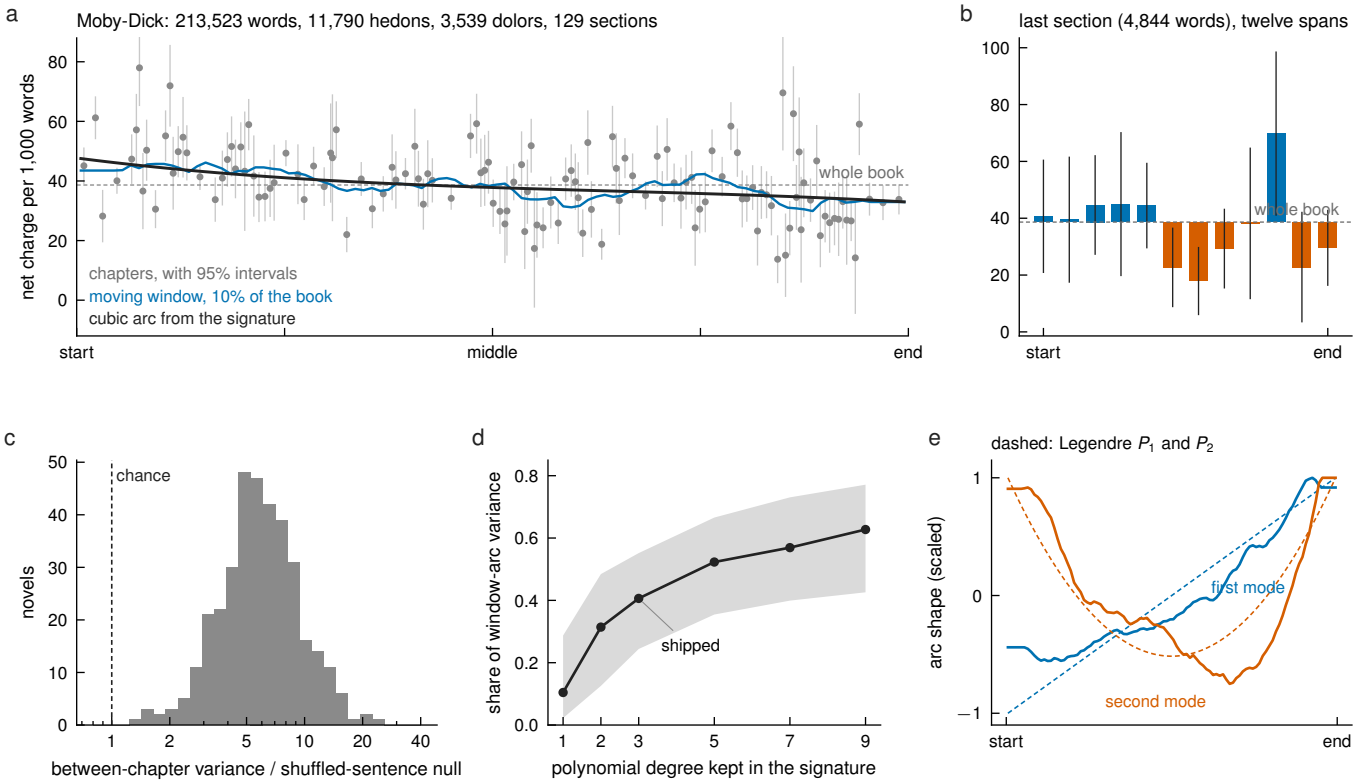


Figure 5 A novel at three scales from one pass. a, *Moby-Dick*: density of each of the 129 sections found by the chapter rule, with 95% intervals from the word-level standard error (too narrow where charge clusters), the cubic arc read from the book's signature, and a moving-window arc for comparison. **b**, The last section (the final chapter and the epilogue) cut into twelve runs of sentences, each read from two rows of the prefix table; bars show the difference from the whole book's density. **c**, Between-chapter variance of density relative to its value when sentences are shuffled across chapters, for 356 novels. **d**, Share of the variance of the moving-window arc captured by the polynomial arc, by the highest degree kept (median and interquartile range). **e**, The two leading modes of variation among the novels' window arcs, with the first two Legendre polynomials.

the cut. It does not make a score built from word ratings a better reader.

How far apart are a pooled value and a mean of parts? For parts with exposures D_p and densities λ_p,

$$\frac{\sum_p D_p \lambda_p}{\sum_p D_p} = \bar{\lambda} + \frac{\mathrm{Cov}(D, \lambda)}{\bar{D}}, \tag{4}$$

an identity familiar from the decomposition of aggregate productivity[48]. The two differ when long parts differ in density from short ones. In novels they rarely do: the median gap between a novel's pooled density and the mean of its chapter densities is 4% of the standard deviation of density between novels, and 3.1% of the 63,190 pairs of novels are ordered differently by the two (Fig. 6d).

A difference between two bodies of text splits in the same spirit. Kitagawa's decomposition[49] writes a difference in pooled density as a part due to differences within strata plus a part due to the mix of strata. Four film critics in the review corpus[40] illustrate it, with the critic's own rating as the stratum (Fig. 6e). Pooled, Berardinelli's and Renshaw's reviews have the same density (difference −0.1 [−1.3, +1.1] per thousand words). Within the same rating Berardinelli's prose is the more negative by −1.2 [−2.5, +0.0], an interval that reaches zero, and his mix of ratings adds back +1.1 [+0.8, +1.6]. None of the 6 pairs of critics reverses between the pooled and the within-rating comparison. The decomposition is bookkeeping, and which part answers a question depends on the question[50]; a signature kept per stratum makes both parts available, and a pooled bounded score does not.

Human and machine writing

A measure that is cheap and does not grow with length can be pointed at a current question: does text written by a language model carry feeling differently from text that people write on the same prompt? Earlier comparisons mostly report that model output is less negative, and more neutral or positive, than human text[51–54], with exceptions by domain[55,56], alongside differences in vocabulary and grammar[57,58]. Before generating or scoring any machine text I fixed ten measures read from signatures and four predictions (Methods): machine text would be more positive overall (HL1), use fewer negative words (HL2), spread its charge more evenly over sentences (HL3), and close on a more positive note than its body (HL4).

Two comparisons were made. HC3 pairs questions with an answer from a person and an answer from ChatGPT[51]; 20,091 pairs from five sources had answers long enough to score. For essays, I sampled 160 essays that US school students wrote on eight assignments[59] and had four Claude models write to each student's assignment and length, once on the bare assignment and once with the instruction to write as a student of that grade would. The models ran as agents inside a coding assistant and were not blind to the project (Methods).

In HC3, pooled over its sources, all four predictions hold by the planned rule, and all four effects are small: ChatGPT answers are higher in net density by +0.14 human standard deviations, lower in dolor density by −0.06, lower in clustering by −0.04 and higher in ending lift by +0.13 (Fig. 7a). Source by source the verdicts differ (Extended Data Table 6). All four hold in 1 of the 5 sources. In the medical questions three of them reverse (net density −0.74, dolor density +0.28, ending lift −0.55), and in the Reddit questions, 70% of

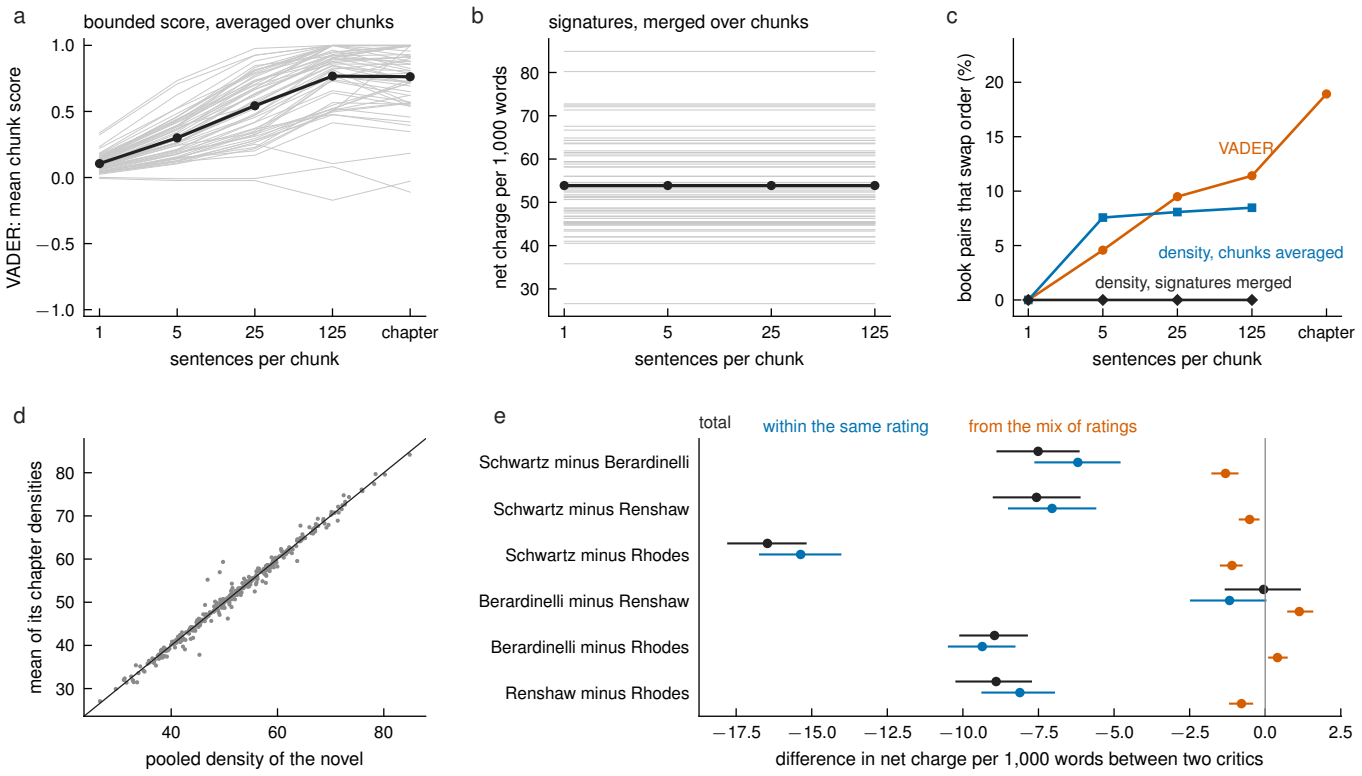


Figure 6 Averaging depends on the cut; merging does not. **a**, Book score for 60 novels computed as the mean VADER compound score over chunks of a given number of sentences (grey, each novel; black, mean). **b**, Density of the same novels from signatures of the same chunks, merged. **c**, Share of pairs of novels whose order changes relative to one-sentence chunks; the series for averaged densities ends at 125 sentences (10% with whole chapters). **d**, Pooled density of each of 356 novels against the mean of its chapter densities (per 1,000 words). **e**, Differences in density between four film critics, split into the part within the same rating and the part due to the mix of ratings (95% intervals).

the pairs, clustering goes the other way (+0.03).

In the essays one prediction holds under both instructions. The final fifth of a machine essay sits 16 per thousand words above the essay's own density, against 3 for the students, a difference of +0.60 [+0.13, +1.07] standard deviations (+0.52 [+0.16, +0.88] when the model imitates a student). The intervals here and below come from a jackknife over the eight assignments, which is wider than the bootstrap over assignments that the plan named (Methods). Overall positivity is undecided. Plain essays are less positive than the students' (−0.23 [−0.48, +0.03]) and essays written as a student more positive (+0.29 [−0.01, +0.60]); the planned bootstrap would have called the first a refutation of HL1 and the second support for it. HL2 and HL3 are undecided for essays.

One difference that I had not predicted is the clearest in the essays. Machine essays put less of their charge directly after an intensifier (−0.58 [−0.90, −0.25] plain, −0.48 [−0.87, −0.09] as a student). Pooled HC3 agrees (−0.21), though the sign is the other way in 2 of its five sources. Charge under a negation is lower in plain essays (−0.40 [−0.73, −0.06]) and in pooled HC3 (−0.26), and undecided for essays written as a student (−0.24 [−0.62, +0.15]). These students wrote *very* next to their feeling words more often than these models did.

The differences are real and modest. A logistic regression on the ten measures tells human from machine text with an area under the curve of 0.63 in HC3, 0.74 [0.68, 0.82] for plain essays and 0.64 [0.56, 0.76] for essays written as a student (Fig. 7b). Word count alone does better in HC3 (0.72), where human answers are shorter (median 102 words against 182), and is close to chance for the essays (0.54 [0.53, 0.56]), which were matched on length to within a few per cent (medians 388 and 404 words). The regression fitted on HC3 scores 0.67 and 0.65 on the essays, point estimates without intervals, so part of the pattern may carry from ChatGPT answers of 2023 to Claude essays of 2026. None of this makes a detector. Purpose-built detectors use far richer evidence and still misfire, notably against non-native writers[60,61].

Can the measures steer a revision? For 57 plain machine essays, the model that wrote each draft revised it under four instructions: a control instruction, to make it read as a student of that grade wrote it; the control instruction plus a report naming each measure on which the draft lay outside the middle 80% of student essays on other assignments; and the control instruction plus one of two published prose skills written to remove the marks of machine writing (Methods). The planned outcome was a text's mean squared standardised distance from the centre of student essays. It turned out to measure how typical a text is, and not how student-like. The student essays themselves average 1.16 on it, because real essays scatter around their own centre, while drafts average 0.66, control revisions 0.53, revisions made with a skill 0.54 and 0.49, and revisions that saw the report 0.41 (Fig. 7c). A smaller distance means a more average essay. On the planned test the report is not distinguishable from the control (difference −0.12 [−0.29, +0.06]), and neither are the skills (+0.01 [−0.09, +0.10] and −0.04 [−0.17, +0.09]). On the secondary outcome, the mean machine score of a classifier trained on other essays, the drafts are at 0.62, the four kinds of revision between 0.53 and 0.56, and the students at 0.38 (Fig. 7d). Every kind of revision moved a little toward the students, and none came close. The report did move what it named. It cut the mean ending lift from +0.31 to +0.02 standard deviations, against +0.21 for the control, and left intensifier use where it was (−0.60 to −0.57), because a draft low on intensifiers is still inside the wide

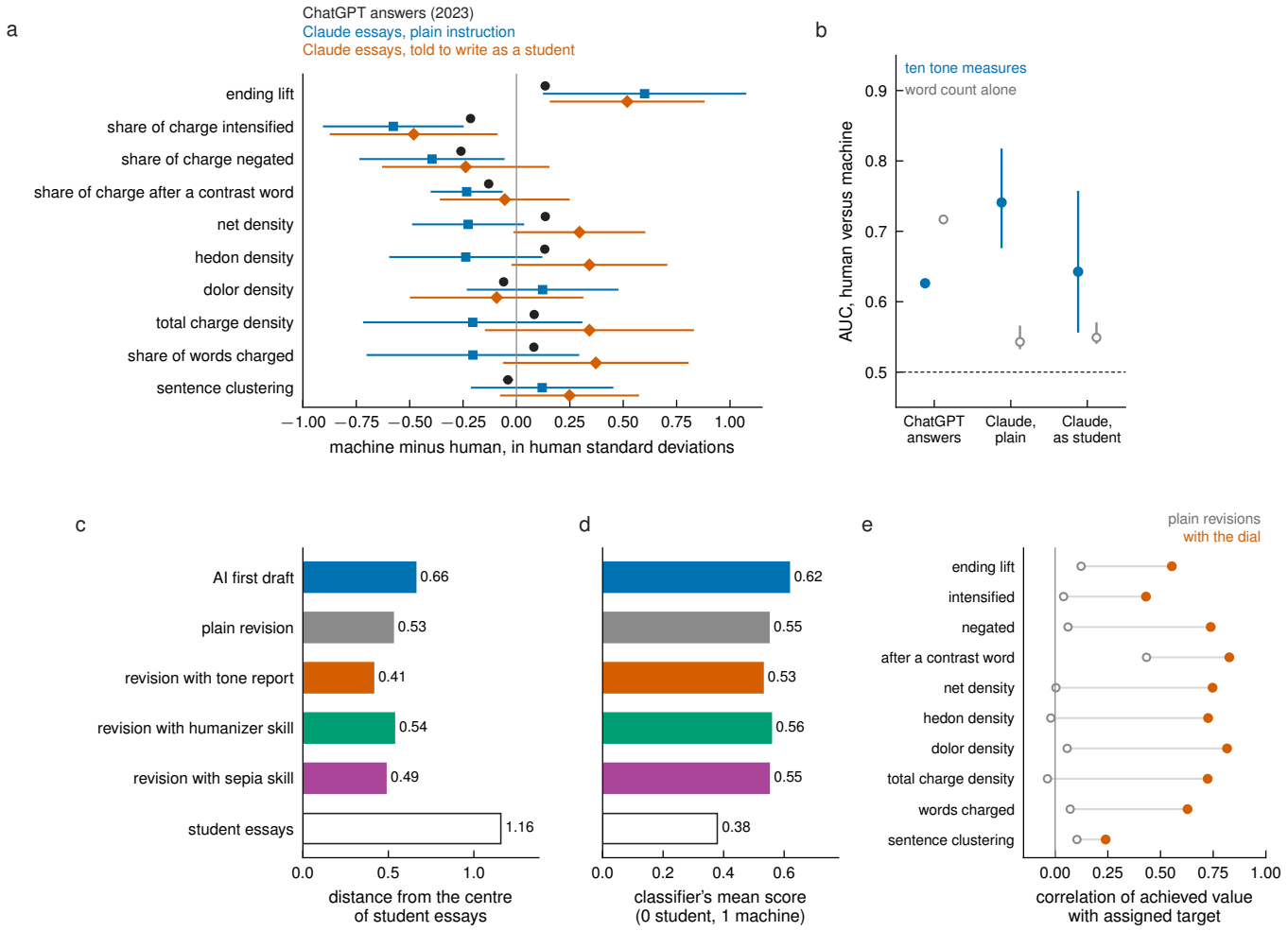


Figure 7 Human and machine writing on the same prompts. a, Difference between machine text and human text on ten measures, in human standard deviations, with 95% intervals: 20,091 questions answered by people and by ChatGPT (bootstrap over questions), and 160 student essays matched by essays from four Claude models under two instructions (jackknife over eight assignments). **b**, Area under the curve of a logistic regression that tells human from machine text using the ten measures (blue) or word count alone (grey), scored out of fold. **c**, Mean squared standardised distance from the centre of student essays for 57 machine drafts, for four kinds of revision of the same drafts, and for the matched student essays themselves, which lie farthest from their own centre. **d**, Mean score of a classifier trained on other essays, from 0 for student essays to 1 for machine essays, for the same texts. **e**, The dial: correlation across 48 essays between the value assigned to a measure and the value achieved after two revisions, with the dial and with plain revisions.

human range and was not flagged.

A report of ranges is a weak instrument. It acts only on drafts outside a wide band, and if it worked perfectly it would pull every draft to one centre. A dial is stronger. Each of 60 further drafts was assigned the measured profile of one real student essay, drawn at random from other assignments, and was revised twice by its author model with a sheet giving the current and the target value of each measure; a control arm revised the same drafts twice with the plain instruction. The smallest of the four models declined 11 of its 15 dial requests, so the result rests on three models and on the 48 essays with both arms complete. The dial was followed. Achieved values correlate with assigned targets at a mean of 0.64 over the ten measures in the dial arm and 0.08 in the control arm (difference +0.56 [+0.30, +0.82]; Fig. 7e), at 0.60 to 0.65 for each of the three models, and from 0.83 for the best-followed measure to 0.24 for sentence clustering. The mean squared gap to the target fell from 1.67 to 1.07 after one round and 0.79 after two, while the control arm stayed at 1.55.

Following the dial did not make the essays more like student writing than a plain revision did. The classifier's mean machine score is 0.64 for the drafts and 0.43 for the matched student essays; two plain revisions bring it to 0.44 and two rounds with the dial to 0.50 (difference +0.06 [−0.11, +0.23], undecided). A model judge read both revisions of each draft in random order without being told which was which, though it was not blind to the study (Methods). It chose the dial revision as the better written in 30 of 48 pairs (exact 95% interval [47, 76]%). That passes the planned test, that the dial revision not be chosen less than half the time, a test lenient enough that passing it says little. The preference came from two of the writers: 14 of 15 pairs for drafts by Opus and 12 of 15 for Fable, against 3 of 14 for Sonnet and 1 of 4 for Haiku. Asked a second question that the plan did not list, which revision reads more like a student's own work, the judge chose the dial revision in 5 of 48 pairs ([3, 23]%), and never on both questions. The dial revisions read as more polished, and polish is not what a student's essay has. Extended Data Table 8 shows the essay at the median improvement, chosen by rule and not by eye. To raise its positive charge the dial revision closes on *a fun, wonderful summer*, and its ending lift moved away from the target.

The measures therefore give a model a working control over how its text carries feeling, as far as a target can be reached by word choice in two passes. That is narrower than writing as a person does. A plain instruction to write as a student did as much by the classifier and more by the judge.

One more observation belongs here. These experiments ask a model to make machine text resemble student writing, and the model sometimes refused: 3 of 240 requests in the first revision experiment, and 11 of 60 essays in the dial arm were left unfinished for that reason, all 11 by the smallest of the four models. The refusals said that passing machine text off as a student's work misrepresents authorship. The tools released with this paper report tone against a human range for writers who disclose their use of a model, and the separations measured above are too weak for these measures to serve as a detector or to defeat one.

Discussion

The aim was a measure of sentiment for which granularity holds. The signature provides one: charge adds, position sums shift, and the signature of a whole is the merge of the signatures of its parts, exactly in integer arithmetic and to rounding error in floating point. Totals, density, a level with a standard error, a predicted rating and an arc are all read from the same nineteen numbers, and the scale at which to read them can be chosen after the text has been scanned.

No component of this is new. Averaged word norms [2], separate positive and negative counts [3,9], mergeable moment summaries [20,21] and the decompositions of pooled values [48,49] have long histories in their own fields, and the description of mergeable position weights is a classical result [29,30]. The paper adds their combination for sentiment, an implementation tested for exactness, and evidence on how position and neutral words enter human judgements.

Of the two hypotheses the work began with, one failed in the form I gave it and one held. Feeling words are not placed early; they sit late in every kind of text examined. Whether early words weigh more with readers is a matter of genre: they do in film-review sentences, and the reverse holds in tweets and in Yelp reviews. My own replacement, recency, passed three tests fixed in advance and did not survive a control for length that those tests should have included. Writing a test down beforehand protects against choosing the analysis after seeing the result. It does not protect against a test that was badly designed, and here a reviewer caught what the plan had missed. Neutral words do dilute, as the averaging account of impression formation predicts [33,34], and they do so from either side. What the phrase ratings support is plainer than either hypothesis: a part's weight in the whole grows with its share of the words, by less than proportion.

The signature is not the most accurate reader of a single text. Transformers agree with raters better on average (0.59 to 0.68 against 0.51), and where the judgement of one text matters they are the better tool. On long documents it is weaker still: on whole film reviews a plain lexicon method beat it, and wherever validity above the sentence could be checked, an average of sentence scores from another method tracked the human target better than the merged signature did. The case for the signature rests on other properties: thousands of times the throughput per processor core, a result that does not change with the way a text is cut, and readouts that can be audited word by word. One sensible division of labour is to scan everything with signatures and to send a larger model only the spans that the scan marks.

Several limits qualify these results. Word norms rate words, not events, so densities of narrative text should be read against the text's own baseline, and irony, sarcasm and meaning carried by syntax beyond the four operators are missed. The lexicon is English and its ratings date from the early 2010s. The constants were fitted on six corpora in which reviews and social media dominate; their readings change between corpora, and charges refined on ratings neither travelled across genres nor stayed neutral about groups of people. No human ratings of chapters or books were available. The standard error of a level treats words as independent draws, which the clustering of charge in sentences and chapters contradicts, so the chapter intervals in Fig. 5a are too narrow, and it is not a prediction interval for a human rating. The chapter rule is approximate. Human judgements of long experiences are known to weigh peaks and endings rather than sums [62]; the signature keeps what an averaging or an end-weighted readout needs, and a readout of peaks would need signatures kept per part. The comparison of human and machine writing covers 160 student essays on eight assignments, four models from one family reached through an agent harness that did not keep them blind, and ChatGPT answers from 2023; the dial rests on three of those models, and its judge was a model, not a human reader. The plan and its addenda were committed to a local repository over one evening and pushed when the study was complete, so their timing rests on the author's record. Analyses added after the first results are marked as such.

Three additions would test the measure further: human ratings of chapters, so that readouts above the sentence can be validated directly; word norms in other languages; and the decayed sum that Proposition 2 allows, for a running level in streams of speech or messages.

Methods

Word charges. labMT 1.0 gives the mean of 50 ratings of happiness on a scale from 1 to 9 for 10,222 frequent English words [2]. The charge of a word is $c = (h - 5)/4$, so that $c = 1$ at the top of the scale, 0 at its neutral point and -1 at the bottom. The VADER list gives the mean of 10 ratings from -4 to 4 for 7,494 words, emoticons and abbreviations [7]. I put it on the labMT scale with one slope through the origin, the ratio of root-mean-square values over the 1,366 shared words, and averaged the two values where both exist. The result has 16,350 entries, of which 8,408 have $|c| \geq 0.25$. Operator words are given no charge, which leaves 8,370 entries that carry charge; the rest are buffer words, as are words outside the list. Positive charge is $h = \max(c, 0)$ hedons and negative charge $d = \max(-c, 0)$ dolors. Charges are stored as integers in thousandths. Adding the NRC-VAD [63] and Warriner [64] norms ($r = 0.88$ and 0.92 with labMT on shared words) raised mean agreement on development data from 0.521 to 0.534; their licences do not allow redistribution, so they are not part of the shipped lexicon and no reported result uses them.

Tokens and operators. Text is lower-cased. Emoji are replaced by their names, using the list distributed with VADER, and 40 emoticons are recognised when they follow a space. Words are runs of letters and apostrophes. A full stop, exclamation mark or question mark ends a sentence; a comma, semicolon or colon ends a clause. Four closed lists of operator words carry no charge themselves: negators (*not*, *never*, forms ending in *n't* and others), intensifiers (*very*, *really*, *so*), diminishers (*slightly*, *somewhat*) and contrast words (*but*, *however*, *yet*, *nevertheless*, *nonetheless*). A negator reaches the next three words or the next punctuation mark, whichever comes first; an intensifier or diminisher reaches the next two words; a contrast word reaches the end of its sentence. Every sentence end clears all operators. The scan is a finite-state transducer and uses string replacement, translation and splitting only.

Signature. For a span of L words with charges h_i, d_i at positions $i = 0, \ldots, L-1$, the signature holds L; the number E of charged words; $S_k^{\pm}$ for $k = 0, \ldots, 3$ (equation 1); the hedons and the dolors inside the reach of each of the four operators; and $\sum_i (h_i - d_i)^2$. The fast implementation computes signatures for a batch of texts with array operations. A reference implementation loops over tokens in integer arithmetic and defines the result; the test suite checks that the two agree, and checks the propositions below for exact equality on random texts.

Proposition 1. Let A and B be texts and let A end at a sentence end. Then $\sigma(AB) = \sigma(A) \oplus \sigma(B)$, where $\oplus$ adds every entry except the position sums and applies equation (2) to those. The operation $\oplus$ is associative, the zero vector is its identity, and for fixed $\sigma(A)$ the map $\sigma(B) \mapsto \sigma(A) \oplus \sigma(B)$ is invertible.

Proof. Operators do not reach across a sentence end, so each word of B has the same charge and the same operator context in AB as in B. Counts and totals therefore add. A word at position j of B is at position $L_A + j$ of AB, and expanding $(L_A + j)^k$ gives equation (2). Write T_ℓ for the linear map that equation (2) applies to the position sums for a shift by ℓ. It is the matrix of the substitution $x \mapsto x + \ell$ on polynomials of degree below four in the monomial basis, so $T_\ell T_m = T_{\ell+m}$ and $T_\ell^{-1} = T_{-\ell}$, which give associativity and invertibility. □

Merging is not commutative: parts can be scored in any order and on any machine, and must be merged in reading order. With prefix signatures $\sigma_j = \sigma(s_1 \cdots s_j)$ over sentences, the signature of sentences i to j is $T_{-L}(\sigma_j - \sigma_{i-1})$ in the position sums, with L the length of the first $i - 1$ sentences. In floating point this difference loses precision for high moments of short spans deep inside long texts; the integer implementation does not.

Proposition 2. Let $\varphi_1, \ldots, \varphi_n$ be linearly independent real sequences on positions $0, 1, 2, \ldots$, and for a real charge sequence c of length L let $\Phi(c) =$

$\big(L, \sum_i c_i\varphi_1(i), \dots, \sum_i c_i\varphi_n(i)\big)$. There is a function F with $\Phi(cc') = F(\Phi(c), \Phi(c'))$ for all charge sequences c, c' if and only if the span V of the φ_r is closed under the shift $(T\varphi)(i) = \varphi(i+1)$. In that case every φ_r satisfies a linear recurrence with constant coefficients of order at most n, and is a combination of sequences $i^k\rho^i$, with ρ a nonzero complex number, and of sequences that vanish from position n onward.

Proof. If V is closed under the shift, then $T^\ell\varphi_r = \sum_s M^{(\ell)}_{rs}\varphi_s$ for some matrices $M^{(\ell)}$, and $\sum_j c'_j\varphi_r(L+j) = \sum_s M^{(L)}_{rs}\sum_j c'_j\varphi_s(j)$, so $\Phi(cc')$ follows from $\Phi(c)$ and $\Phi(c')$. Conversely, fix $\ell \geq 1$ and let c be ℓ zeros. Take J large enough that the restrictions of the φ_s to positions below J are linearly independent, and let c' range over sequences of length J. By assumption $\sum_j c'_j\varphi_r(\ell+j)$ depends on c' only through the numbers $\sum_j c'_j\varphi_s(j)$. Adding to c' any d with $\sum_j d_j\varphi_s(j) = 0$ for all s therefore leaves it unchanged, so the linear functional $d \mapsto \sum_j d_j\varphi_r(\ell+j)$ vanishes on the common kernel of the n functionals $d \mapsto \sum_j d_j\varphi_s(j)$ and is a linear combination of them. The coefficients are unique, and do not change with J because the equations for J are among those for $J+1$. Hence $T^\ell\varphi_r \in V$. The shift is then a linear map on the n-dimensional space V and, by the Cayley–Hamilton theorem, is annihilated by its characteristic polynomial p, so every member of V satisfies the recurrence $p(T)\varphi = 0$. Its solutions are spanned by $i^k\rho^i$ for the nonzero roots ρ of p and, for a zero root of multiplicity m, by the sequences supported on positions below m. □

The proposition concerns summaries that are linear in the charges. It is the sequence form of the description of finite-dimensional translation-invariant function spaces [29,30], and of the fact that a function on strings has a finite-dimensional linear representation exactly when its Hankel matrix has finite rank [31,32]. A geometric weight ρ^i gives the one-number case $S_\rho(AB) = S_\rho(A) + \rho^{L_A}S_\rho(B)$. On relative position: a weight $g\big((i+\frac{1}{2})/L\big)$ is available from a summary for every length L only if the span of weights is closed under rescaling of position as well as under shifts. A finite-dimensional space of functions of a real variable with both properties is the space of polynomials up to some degree: closure under shifts gives exponential polynomials [29], rescaling rules out any nonzero exponent λ because the functions $e^{\lambda s x}$, $s > 0$, are linearly independent, and a shift-invariant space of polynomials contains the derivatives of its members.

Readouts. Density is $1000\,(H-D)/L$. Level is equation (3); its standard error is $1000\sqrt{\sum_i (h_i - d_i)^2 - (H-D)^2/L}\,/\,(a + E + \kappa B)$, which treats the charges of a text's words as independent draws. With $x_i = (2i+1)/L - 1$ and Legendre polynomials P_k, the moments $m_k = \sum_i (h_i - d_i)P_k(x_i)$, $k \leq 3$, are linear combinations of the position sums, and the arc is $f(x) = \sum_k \frac{2k+1}{L} m_k P_k(x)$. The sampled polynomials are orthogonal up to terms of order L^{-2}, so f is the least-squares cubic to that order; over the novels it agrees with a direct fit to 10^{-8}. The calibrated valence is $b_0 + \mathbf{b}\cdot\mathbf{x}/(a+E+\kappa B)$, where $\mathbf{x}$ holds H, D, the moments m_1, m_2, m_3, the negated hedons, the negated dolors, and the net charge that is intensified, diminished, and after a contrast word.

Corpora. Table 1 lists the rated corpora. Official training, development and test portions were kept where they exist. Facebook posts and film reviews were split 40/10/50 by a seeded shuffle. Yelp training and development portions are seeded samples of the official training set.

Table 1 Rated corpora. Numbers of texts in the training, development and test portions.

Corpus	Unit	Human rating	Train / dev / test
EmoBank [36]	sentence, six genres	valence, 1 to 5	8,062 / 1,000 / 1,000
SemEval-2018 [37]	tweet	valence, 0 to 1	1,181 / 449 / 937
Facebook posts [38]	post	valence, 1 to 9	1,158 / 289 / 1,448
SST [39]	sentence	sentiment, 0 to 1	8,544 / 1,101 / 2,210
Film reviews [40]	review	critic's rating	2,002 / 501 / 2,503
Yelp [41]	review	stars, 1 to 5	100,000 / 10,000 / 50,000

Fitting and evaluation. Ratings were standardised within each corpus with the mean and standard deviation of its training portion, and corpora were given equal total weight. For each target corpus and each pair (a, κ) on the grid $a \in \{0, 0.5, 1, 2, 4, 8, 16, 32\}$, $\kappa \in \{0, 0.05, 0.1, 0.2, 0.5, 1, 2\}$, the coefficients were fitted by ridge regression on the training portions of the other five corpora (penalty 10^{-3} times the mean diagonal of the weighted Gram matrix; intercept unpenalised). The pair with the highest mean Pearson r on those corpora's development portions was kept, the coefficients were refitted on training plus development portions, and the target's test portion was scored once. The shipped constants ($a = 4$, $\kappa = 0.5$) are the same fit on all six corpora. Intervals are 95% percentile bootstrap intervals, drawing source documents for EmoBank and items elsewhere; differences between methods use the same resamples. Pearson r and differences between methods use 2,000 resamples, as planned. To bound running time Spearman's ρ uses 500, and the position, dilution, treebank and machine-writing analyses use between 100 and 1,000; the scripts give each count. The interval for the mean difference from VADER is a t interval over the five primary corpora. Comparators: VADER compound score; the hedonometer, the mean labMT value of words outside its neutral band [2]; TextBlob polarity; ridge regression on TF-IDF weights of words and word pairs, trained on the target's own training portion or on the other five corpora; and two transformer classifiers scored as $P(\text{positive}) - P(\text{negative})$ on text truncated to 512 tokens, DistilBERT fine-tuned on SST-2 [12] and a RoBERTa model fine-tuned on tweets [42]. The transformers were run on a seeded sample of 5,000 Yelp test reviews, and for Yelp all methods are compared on that sample.

Refined charges. With exposure $a + L$ and operator multipliers θ from a one-channel fit, the predicted rating of text t is $b_0 + s\sum_i \mu_i c_{w_i}/(a + L_t)$ with $\mu_i = 1 + \theta\cdot$(operator flags of word i), which is linear in the vector c of word charges. The refined charges minimise the weighted squared error plus $\lambda\|s(c - c^0)\|^2$, where c^0 holds the norm charges and words that occur in at least 25 training texts start at zero. The penalty $\lambda \in \{0.03, 0.1, 0.3\}$ was chosen on development data of the fitting corpora, and the readout was then refitted with the refined charges. Each step is a ridge regression; the sequence is not a joint optimum.

Position and dilution. For where charge sits (planned, H1a), each text of at least ten words contributes $\sum_i |c_i|x_i$ and $\sum_i |c_i|$, and the statistic is the ratio of their sums over texts. Besides the six rated corpora, the units were paragraphs of the film reviews and chapters of 150 of the novels, so the eight kinds of unit come from seven bodies of text. The position kernel of a corpus (planned, H1b) is $1 + (b_1P_1 + b_2P_2 + b_3P_3)/b_0$ from a one-channel fit to all its portions with the shipped a and κ; the contrast is its mean over the first third minus its mean over the last third. For EmoBank documents of eight or more sentences (planned, H1c), $|V - 3|$ was regressed on relative sentence position with document means removed, resampling documents. The planned test of dilution (H2) is the paired difference in held-out r between the calibrated readout and the same readout refitted with $\kappa = 0$.

The remaining analyses were not in the plan. For dilution in the corpora, rated intensity is the absolute deviation of a rating from the corpus median, and the partial rank correlation with the number of buffer words removes the rank of total charge from both by linear regression. The curves of agreement against κ are fits to single corpora with a fixed at the shipped value, on a grid extended to $\kappa = 4$. For the treebank, every binary node whose phrase and both parts have ratings in the phrase dictionary was used (100% of nodes in the training trees); ratings were centred at 0.5, the phrase rating was regressed on the two part ratings, and sentences were resampled. With length held fixed, nodes whose parts both have two or more words were split by which part is longer, and the weights were also fitted as one step function of a part's share of the phrase's words in ten bins, shared by both parts, plus a constant for the later part; a second fit gives each part its own step function (Fig. 3b). For dilution in the treebank, a part counts as rated if its centred rating is at least 0.15 in absolute value and as neutral if at most 0.05; the share kept is the ratio of sums of the signed phrase rating to the absolute part rating. The matched comparison of sides weights cells defined by the length of the rated part and the length of the neutral part (1, 2, 3 to 4, 5 to 8, 9 or more words each) by the smaller of their two counts. These analyses are reported for all trees, with the development and test trees alone where the recency tests are discussed.

Novels, fairy tales and critics. From a seeded sample of English fiction in the Project Gutenberg catalogue, 640 texts were fetched from a mirror, and 356 had at least ten chapters of 300 or more words and 20,000 words in all after removal of front matter. Chapters were found by headings, sections under 300 words were dropped, and each chapter was closed with a full stop. The rule is approximate. In *Moby-Dick* it finds 129 sections for the book's 135 chapters and epilogue: it drops the chapters shorter than 300 words, joins the epilogue to the last chapter, and keeps one block of transcriber's notes as a section. The null for chapter differences shuffles a novel's sentences 200 times and keeps the number of sentences in each chapter, so the smallest attainable p is $1/201$. The comparison of chunk sizes with VADER used the first 60 novels in file order. Arcs were compared on 100 points; the moving window is 10% of the words. The fairy-tale annotations [47] give two annotators' emotion labels for each sentence; a label of happy or positively surprised was scored +1, neutral 0, and any other emotion −1, and the two were averaged. Critic comparisons use all 5,006 reviews with the four-level rating as stratum; intervals resample reviews within critic.

Human and machine writing. The ten measures are: hedons, dolors, net charge and total charge per thousand words; the percentage of words that are charged; the percentage of charge that is negated, intensified, and after a contrast word; ending lift, the mean of the cubic arc over the last fifth of the text minus the text's density; and sentence clustering, the between-sentence variance of density divided by its expected value if the text's words were dealt to its sentences at random, $\sigma^2(m-1)/L$ for m sentences, L words and word-level variance σ^2 of net charge. None grows with the length of a text, though all are noisier in short ones. From HC3 [51] I took the first human answer and the first ChatGPT answer to each question and kept the 20,091 questions where both have at least 30 words; contractions split by the corpus's tokenisation of human answers were rejoined. From PERSUADE 2.0 [59] I drew 20 essays of 150 to 800 words for each of the eight assignments that need no source text (160 essays, seed 0). Each essay was matched by two machine essays written to the same assignment and word count, one with no further instruction and one with the instruction to write as a student of the same grade would in a timed test. Two of the 320 requests produced no essay, both under the plain instruction.

The writers were subagents of the coding assistant, one essay per agent, running Claude Fable 5.2, Opus 5.5, Sonnet 5.5 and Haiku 5.5 in rotation at low reasoning effort. Each agent's prompt held its task and a file path to write to. The harness also attached its standing context to every agent: an index of the author's project notes, a coding-style plug-in and the author's most recent message to the assistant. For the essays that message described the planned comparison of machine and human essays; for the dial revisions and the judges it stated the author's hope that steering by these measures would make machine text read as more human. The file paths named the experimental arm. The agents were therefore not blind to the project, and the revisers and the judge were not blind to the hoped-for result. The job files and the workflow scripts that dispatched them, with the judge's instruction, are in the repository; the agents' transcripts are not.

Differences are paired by essay or question and standardised by the human standard deviation. HC3 intervals are percentile bootstrap intervals over questions. For the essays the plan named a bootstrap over the eight assignments. A percentile bootstrap over eight

clusters is too narrow, so the printed intervals are from a delete-one-assignment jackknife with a t reference on seven degrees of freedom, and the planned verdicts are stated where they differ. Resampling essays, HL3 is refuted for essays written as a student, which cluster more than the students' and not less. The classifier is a logistic regression on the ten standardised measures, scored out of fold with all texts of an assignment or question in one fold; its intervals are bootstrap intervals over assignments or questions.

For the tone check, 60 plain-instruction drafts, 15 per model drawn with one seed, were each revised by the model that wrote them under four instructions: the control instruction, to revise the draft so that it reads as a student of that grade wrote it; the control instruction plus a report listing each measure on which the draft lay outside the middle 80% of student essays on other assignments; and the control instruction plus an instruction to do the rewrite with one of two published prose skills for removing the marks of machine writing (*humanizer* and *sepia-recreate*). Distance is the mean over measures of the squared standardised difference from the mean of student essays on other assignments. The model declined 3 of the 240 revision requests, stating that making machine text read as a student's work would misrepresent authorship; essays with all four revisions (57, from 7 of the eight assignments) enter the comparison. The dial used 60 further drafts drawn the same way and also covers 7 assignments. Its classifier was trained on the 98 essay pairs outside that sample. The judge was Claude Opus 5.5, or Sonnet 5.5 for drafts written by Opus; shares chosen by the judge are given with exact binomial intervals that treat pairs as independent.

Throughput. Timings were taken on a rented machine (AMD Ryzen 5 5600X 6-Core Processor, six cores and twelve threads; RTX 3090) so that processor and graphics-card figures come from one host. The plan had named the author's laptop, which was replaced because long runs overheated it. The texts are 200,000 sentences from Yelp reviews (mean 16 words). Each method scored a prefix of the same list: all of it for the signature, 20,000 sentences for the hedonometer, 5,000 for VADER and TextBlob, 600 for each transformer on one thread, 3,000 on twelve threads and 20,000 on the graphics card (batches of 128 sorted by length, single and half precision). The timed operation for the signature is the scan to signatures followed by the level readout; the calibrated readout takes about 6% longer, measured afterwards on the author's laptop. Signature timings are the best of three runs and lexicon timings the best of two; each transformer was timed once. The host ran PyTorch 2.3.1.

Pre-specification. The method, the corpora, the outcomes, the comparators and the decision rules for the hypotheses were written into an analysis plan and committed to the project's git repository (commit `7524c58`) before any evaluation data were scored. Four addenda were committed before the data they concern were generated or examined: the comparison of human and machine writing (`4b10b12`), the recency hypothesis and its three tests (`77d4fa2`), the rewrites with prose skills (`090411d`) and the dial (`de181ec`). This is a record kept by the author and not an independent registration. The commits were made in a local repository over one evening, they carry the timestamps of the author's machine, and they were pushed to the public repository only when the study was complete. The order of file times agrees with the stated order, with one exception: the note that narrows the essay sample to eight assignments was committed together with the second addendum, after the essays had been written.

Test portions were first scored with the plan as written; the results of that first run were not kept, so the statement that the changes below left the accuracy criterion met cannot be checked from the repository. Deviations, all listed in the plan: the grid for a and κ was widened after the first run chose its upper edge, without changing the choice; the tokeniser was rewritten for speed; a bug in the selection of constants for the uncalibrated level was fixed; the VADER list was put on the labMT scale by a ratio of root-mean-square values where the plan said regression; several bootstraps use fewer resamples than the planned 2,000; the essay intervals are jackknife intervals; the planned report of how often the document score and the mean of sentence scores differ in sign was at first made for the transformers only, and VADER was added later; and the classifier for the dial was first trained on a set that left out only the essays with both arms complete, then retrained leaving out all 60 sampled essays. Refined charges, the bag-of-words ridge trained on other corpora, the partial correlations, the curves against κ, every treebank analysis, the readings of the constants, the check of error against exposure and the second question to the judge were added after the first results and are exploratory. The length-controlled treebank analysis, the jackknife intervals, the table of HC3 by source and the check of refined charges for words about groups were added in response to reports by referee agents. Where a model declined a request, the text is missing and the essay leaves the paired comparison; no declined request was retried. A February 2026 version of the method, built on word embeddings, failed its own criterion of matching VADER on GoEmotions; the criterion carried over here is a mean r over the primary corpora at least that of VADER.

Software. Python 3.10, NumPy 2.0, SciPy 1.15, scikit-learn 1.5, PyTorch 2.11 and Transformers 4.51 on the author's machine; the timings used PyTorch 2.3.1 on the rented host. Figures were drawn with Matplotlib 3.9 and SciencePlots [65].

Use of AI tools. The aims and the two hypotheses come from the author's 2024 manuscript. The code, the analyses and the text of this draft were produced with Claude (Anthropic), an AI assistant working in a coding environment under the author's direction, and were reviewed by AI referee agents whose reports are in the repository under `reviews/`. One of those reports found the confound that led to the withdrawal of the recency hypothesis. The essays in the comparison of human and machine writing were written by Claude models by design. Responsibility for the content rests with the author.

Data availability

All corpora and lexicons are public and are downloaded by `scripts/fetch_data.py`. The repository redistributes the lexicon built from labMT (CC BY) and the VADER word list (MIT), VADER's emoji list, summary statistics of the reference texts, the machine-written essays, short excerpts of PERSUADE essays in a comparison page, the result files behind every number and figure, and the analysis plan; licences and attributions are in `THIRD_PARTY_NOTICES.md`. The folder `legacy/2024` keeps the scripts and data of the author's 2024 manuscript and is not used by this study.

Code availability

`https://github.com/ayazkhan27/anonymity`. The numbers in the text are generated by `scripts/make_numbers.py` from the result files, the released lexicon and, for the worked example and one count, the downloaded data.

Declarations

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Extended data

Extended Data Table 1 Pearson r with human ratings on held-out test portions, with 95% bootstrap intervals. The mean is over the five corpora with continuous human ratings (all but Yelp). Signature constants are fitted on the other five corpora unless stated.

	EmoBank	Tweets	Facebook	SST	Film reviews	Yelp	Mean
n (test portion)	1,000	937	1,448	2,210	2,503	5,000	
Signature, calibrated	0.48 [0.43, 0.54]	0.70 [0.67, 0.73]	0.62 [0.58, 0.65]	0.44 [0.40, 0.47]	0.29 [0.26, 0.33]	0.61 [0.59, 0.62]	0.51
Signature, level only	0.44 [0.39, 0.49]	0.71 [0.67, 0.74]	0.57 [0.54, 0.60]	0.42 [0.39, 0.46]	0.26 [0.22, 0.29]	0.54 [0.52, 0.56]	0.48
Signature, constants from the same corpus	0.48 [0.43, 0.55]	0.73 [0.70, 0.76]	0.62 [0.58, 0.65]	0.46 [0.43, 0.49]	0.40 [0.37, 0.44]	0.63 [0.61, 0.64]	0.54
Signature, refined charges	0.41 [0.36, 0.46]	0.71 [0.68, 0.74]	0.58 [0.55, 0.61]	0.55 [0.52, 0.58]	0.49 [0.47, 0.52]	0.64 [0.62, 0.65]	0.55
VADER	0.46 [0.41, 0.50]	0.71 [0.68, 0.75]	0.58 [0.54, 0.62]	0.45 [0.42, 0.48]	0.25 [0.21, 0.29]	0.55 [0.54, 0.57]	0.49
Hedonometer	0.44 [0.40, 0.48]	0.59 [0.55, 0.63]	0.49 [0.45, 0.52]	0.34 [0.30, 0.37]	0.24 [0.20, 0.28]	0.54 [0.52, 0.56]	0.42
TextBlob	0.31 [0.24, 0.36]	0.51 [0.46, 0.56]	0.53 [0.49, 0.57]	0.41 [0.37, 0.44]	0.46 [0.43, 0.49]	0.60 [0.58, 0.61]	0.44
Bag-of-words ridge, other corpora	0.32 [0.27, 0.40]	0.64 [0.60, 0.67]	0.58 [0.54, 0.61]	0.56 [0.53, 0.58]	0.64 [0.61, 0.66]	0.67 [0.65, 0.68]	0.55
Bag-of-words ridge, same corpus	0.51 [0.47, 0.55]	0.59 [0.55, 0.63]	0.53 [0.50, 0.57]	0.66 [0.64, 0.68]	0.77 [0.75, 0.78]	0.83 [0.83, 0.84]	0.61
DistilBERT (SST-2)	0.50 [0.46, 0.53]	0.60 [0.56, 0.64]	0.54 [0.50, 0.57]	0.78 [0.76, 0.79]	0.56 [0.53, 0.58]	0.75 [0.73, 0.76]	0.59
Twitter-RoBERTa	0.73 [0.70, 0.76]	0.81 [0.78, 0.83]	0.78 [0.76, 0.80]	0.77 [0.75, 0.79]	0.31 [0.27, 0.34]	0.76 [0.75, 0.77]	0.68

Extended Data Table 2 Spearman's ρ with human ratings, the plan's secondary outcome, laid out as Extended Data Table 1 (500 resamples).

	EmoBank	Tweets	Facebook	SST	Film reviews	Yelp	Mean
Signature, calibrated	0.48 [0.43, 0.53]	0.71 [0.67, 0.74]	0.60 [0.56, 0.63]	0.44 [0.40, 0.47]	0.30 [0.26, 0.33]	0.64 [0.63, 0.66]	0.51
Signature, level only	0.43 [0.37, 0.48]	0.72 [0.68, 0.74]	0.58 [0.54, 0.61]	0.42 [0.38, 0.45]	0.26 [0.23, 0.30]	0.59 [0.56, 0.60]	0.48
Signature, constants from the same corpus	0.50 [0.45, 0.54]	0.74 [0.70, 0.76]	0.61 [0.57, 0.64]	0.45 [0.41, 0.48]	0.40 [0.37, 0.43]	0.65 [0.63, 0.67]	0.54
Signature, refined charges	0.41 [0.36, 0.47]	0.71 [0.68, 0.74]	0.60 [0.57, 0.63]	0.55 [0.52, 0.58]	0.50 [0.47, 0.53]	0.69 [0.67, 0.70]	0.55
VADER	0.45 [0.40, 0.51]	0.72 [0.69, 0.75]	0.59 [0.55, 0.62]	0.45 [0.42, 0.48]	0.33 [0.29, 0.36]	0.55 [0.53, 0.57]	0.51
Hedonometer	0.44 [0.39, 0.49]	0.62 [0.57, 0.66]	0.50 [0.46, 0.54]	0.34 [0.30, 0.38]	0.25 [0.21, 0.28]	0.60 [0.58, 0.62]	0.43
TextBlob	0.28 [0.20, 0.33]	0.51 [0.46, 0.56]	0.54 [0.49, 0.57]	0.42 [0.38, 0.45]	0.45 [0.42, 0.49]	0.64 [0.62, 0.65]	0.44
Bag-of-words ridge, other corpora	0.31 [0.25, 0.39]	0.63 [0.59, 0.67]	0.58 [0.54, 0.61]	0.56 [0.53, 0.59]	0.64 [0.62, 0.67]	0.68 [0.67, 0.70]	0.55
Bag-of-words ridge, same corpus	0.51 [0.46, 0.56]	0.58 [0.53, 0.62]	0.53 [0.49, 0.57]	0.67 [0.64, 0.69]	0.78 [0.76, 0.80]	0.85 [0.84, 0.85]	0.61
DistilBERT (SST-2)	0.61 [0.56, 0.63]	0.67 [0.63, 0.71]	0.62 [0.59, 0.65]	0.81 [0.80, 0.82]	0.62 [0.60, 0.65]	0.79 [0.78, 0.80]	0.67
Twitter-RoBERTa	0.75 [0.71, 0.78]	0.84 [0.81, 0.85]	0.83 [0.80, 0.84]	0.76 [0.74, 0.78]	0.30 [0.26, 0.33]	0.78 [0.77, 0.79]	0.70

Extended Data Table 3 Charge-weighted mean position of words, from -1 (first word) to $+1$ (last word), for texts of ten or more words, with 95% intervals; and the same for hedons and dolors separately.

Unit	Texts	All charge	Hedons	Dolors
EmoBank	5,975	+0.060 [+0.052, +0.068]	+0.051	+0.107
SemEval-2018 tweets	2,127	+0.105 [+0.096, +0.114]	+0.090	+0.156
Facebook posts	1,698	+0.021 [+0.012, +0.031]	+0.018	+0.036
SST sentences	9,358	+0.050 [+0.046, +0.055]	+0.042	+0.087
Film reviews	5,006	+0.023 [+0.022, +0.025]	+0.024	+0.022
Yelp reviews	157,452	+0.002 [+0.001, +0.002]	−0.001	+0.023
Film review paragraphs	22,514	+0.017 [+0.016, +0.019]	+0.011	+0.041
Novel chapters	2,583	+0.007 [+0.005, +0.009]	+0.005	+0.015

Extended Data Table 4 Coefficients of the shipped calibrated readout ($a = 4$, $\kappa = 0.5$), fitted on six corpora. The readout predicts a human valence rating in standard-deviation units.

Signature entry (per unit exposure)	Coefficient
intercept	−0.356
hedons	+5.850
dolors	−9.554
net charge $\times P_1$	+0.880
net charge $\times P_2$	+1.329
net charge $\times P_3$	+1.259
negated hedons	−15.196
negated dolors	+9.643
intensified net charge	+7.817
diminished net charge	−2.533
net charge after contrast	−0.470

Extended Data Table 5 Readings of the fitted coefficients, by fit. Dolor / hedon: how far a dolor moves the predicted rating relative to a hedon. Negated hedon, negated dolor: the multiple of its charge that a negated word counts for. Intensifier: the charge added, as a multiple of the modified word's. The fit to film reviews alone is poorly determined.

Fit	Dolor / hedon	Negated hedon	Negated dolor	Intensifier
Shipped fit, six corpora ($a = 4$, $\kappa = 0.5$)	+1.63	−1.60	−0.01	+1.02
Other five corpora, EmoBank left out ($a = 2$, $\kappa = 0.2$)	+1.17	−1.19	−0.14	+1.08
Other five corpora, Tweets left out ($a = 4$, $\kappa = 0.5$)	+1.81	−1.95	−0.11	+1.06
Other five corpora, Facebook left out ($a = 4$, $\kappa = 0.5$)	+1.64	−1.79	−0.03	+0.99
Other five corpora, SST left out ($a = 4$, $\kappa = 0.5$)	+1.75	−1.75	+0.06	+1.09
Other five corpora, Film reviews left out ($a = 4$, $\kappa = 0.5$)	+1.63	−1.54	−0.02	+0.96
Other five corpora, Yelp left out ($a = 4$, $\kappa = 0.5$)	+1.57	−1.10	+0.10	+0.87
EmoBank alone ($a = 16$, $\kappa = 2$)	+2.22	−1.75	+0.19	+0.51
Tweets alone ($a = 2$, $\kappa = 0.5$)	+1.28	−0.51	+0.34	+0.93
Facebook alone ($a = 4$, $\kappa = 0.5$)	+1.64	−0.78	+0.07	+1.05
SST alone ($a = 1$, $\kappa = 0.5$)	+0.81	−0.69	−0.78	+0.66
Film reviews alone ($a = 32$, $\kappa = 0$)	−0.25	−1.23	+1.32	+7.89
Yelp alone ($a = 32$, $\kappa = 1$)	+4.48	−4.26	−0.39	+1.10

Extended Data Table 6 ChatGPT answers minus human answers in HC3, by source, in human standard deviations. For the four planned predictions: no mark, supported in that source; †, refuted; ○, undecided (bootstrap over questions). The last two columns were not predicted.

Source	Pairs	Net (HL1)	Dolors (HL2)	Clustering (HL3)	Ending lift (HL4)	Negated	Intensified
Reddit, explain like I'm five	14,009	+0.22	−0.10	+0.03†	+0.24	−0.24	−0.23
Finance	3,664	+0.13	−0.00○	−0.13	+0.08	−0.52	−0.31
Medicine	1,155	−0.74†	+0.28†	−0.45	−0.55†	−0.11	−0.07
Wikipedia, computer science	824	+0.14	−0.13	−0.10	+0.34	+0.03	+0.11
Open questions	439	+0.03○	−0.01○	+0.30○	+0.08○	+0.12	+0.10
All sources	20,091	+0.14	−0.06	−0.04	+0.13	−0.26	−0.21

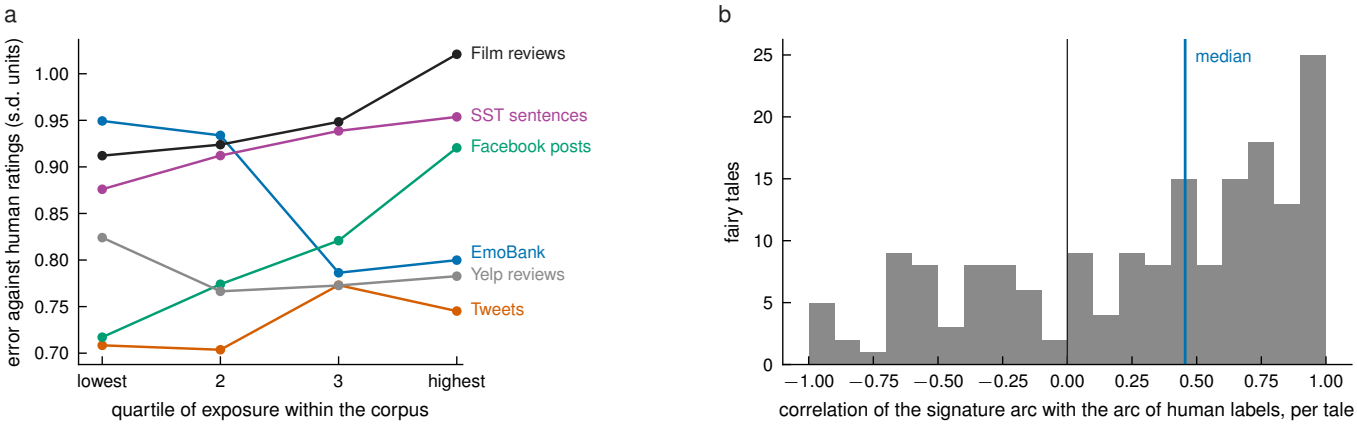
Extended Data Table 7 Charge of 20 words about groups of people and public figures in the human word norms and after refinement on rated text (shipped refinement, all six corpora). The scale runs from -1 to $+1$.

Word	Norm	Refined	Word	Norm	Refined
women	+0.53	−0.08	*black*	−0.03	−0.07
men	+0.23	+0.09	*white*	+0.32	+0.14
girls	+0.48	+0.01	*asian*	+0.33	+0.01
boys	+0.29	+0.09	*mexican*	+0.10	+0.10
gay	−0.14	−0.39	*immigrants*	+0.06	+0.21
lesbian	+0.04	+0.03	*obama*	+0.23	−0.53
muslim	−0.09	−0.37	*trump*	+0.07	−0.14
islam	−0.08	−0.39	*old*	−0.26	−0.30
christian	+0.33	+0.25	*disabled*	−0.55	−0.63
jewish	+0.18	+0.26	*poor*	−0.61	−1.00

Extended Data Table 8 The dial at work on one essay, chosen as the essay at the median improvement in gap to its target (model: Sonnet; assignment: Seeking multiple opinions). First and last 45 words of the machine draft, of its revision after two rounds with the dial, and of its revision after two plain rounds; and five of the measures with the assigned target.

AI first draft	After two rounds with the dial	After two plain revisions
Opening. When I had to choose between two summer programs last year, my first instinct was to ask my mom and go with whatever she said. She thought the robotics camp was a safe bet because it looked good for college. But then I talked to . . . *Closing.* . . . it. But asking a few trusted people, especially people with different experiences, gives you more information, exposes hidden bias, and sharpens your own thinking. That is why I would always rather ask several people than rely on one, even when the first answer sounds convincing.	*Opening.* When I had to choose between two summer programs last year, my first idea was to ask my mom and do whatever she said. She thought the robotics camp was a safe choice because it would look great for college. Then I talked to my . . . *Closing.* . . . the end the choice belongs to the person who has to live with it. I feel glad I did not ask just one person. Kids who ask a few trusted people get a fair chance to choose, and that makes for a fun, wonderful summer.	*Opening.* Last summer I had to choose between two summer programs. At first I just wanted to ask my mom and do whatever she said. She told me the robotics camp was the safe choice because it would look good on my college applications. But then . . . *Closing.* . . . you ask a few people you trust, especially people who have had different experiences, you get more information and you can spot bias. That is why I would always ask a few people instead of just one, even if the first answer sounds really good.

Measure	Draft	Target	Dial	Plain
Hedons per 1,000 words	90.1	105.6	113.5	94.3
Dolors per 1,000 words	4.8	15.1	6.3	4.5
Charge negated (%)	4.7	7.3	7.6	5.7
Charge intensified (%)	0.8	2.0	0.0	3.7
Ending lift (per 1,000 words)	8.2	−19.7	12.7	−0.2



Extended Data Fig. 1 Error against exposure, and arcs of fairy tales. a, Root-mean-square error of the calibrated score against human ratings on each test portion, in standard-deviation units, by quartile of exposure within the corpus. **b**, Correlation between the cubic arc of a tale's merged signature and the cubic arc of its sentence labels from two human annotators, for 176 fairy tales.